Study Buddy STEM AI: An AI-Powered Personalized, Adaptive and Multimodal STEM Learning Ecosystem for Sri Lankan Students

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Abstract

The increasing complexity of STEM education and the diversity of learner needs create a demand for educational systems that can provide personalized, adaptive, and multimodal learning support. Conventional STEM learning environments commonly rely on textbook-based instruction, one-way teaching, and standardized assessments, providing limited opportunities for individualized tutoring and continuous learning support. This paper presents Study Buddy STEM AI, an integrated AI-powered STEM learning ecosystem developed for the Sri Lankan General Certificate of Education Ordinary Level (GCE O/L) context. The system integrates four independently developed but interconnected components: RAG-Based Personalized STEM Storytelling, Multilingual Conversational AI, Emotion-Aware Adaptive Assessment, and Handwritten Content Analysis & Synthesis. The platform combines Retrieval-Augmented Generation (RAG), machine learning, large language models, speech processing, facial-expression recognition, optical character recognition, semantic retrieval, and knowledge synthesis.

Index Terms— Artificial intelligence in education, STEM education, GCE O/L, personalized learning, adaptive assessment, Retrieval-Augmented Generation

I. INTRODUCTION

STEM education plays an important role in developing scientific reasoning, analytical thinking, problem-solving, and technological skills. However, conventional school-based STEM learning frequently depends on textbook-oriented instruction, teacher-led explanations, and standardized assessments. Within the Sri Lankan GCE O/L context, these approaches can provide limited opportunities for individualized learning, interactive concept exploration, adaptive assessment, and personalized revision support. The Study Buddy project identified challenges including low engagement, difficulty understanding abstract scientific concepts, limited adaptive learning support, assessment-related stress, poor knowledge retention, and limited access to personalized tutoring and revision assistance. Recent research demonstrates that artificial intelligence can support personalized STEM learning by adapting content, providing intelligent tutoring, generating feedback, and analyzing learner behavior [1]–[4]. A systematic review of personalized STEM education identified AI-supported personalization as an important direction across K–12 learning environments [1]. Recent work on AI in STEM education additionally emphasizes adaptive assessment, student agency, and the need for responsible integration of AI within educational ecosystems [2]. Adaptive intelligent tutoring systems have also demonstrated the potential of dynamically personalized feedback for STEM learning [3].

Another important limitation of existing educational AI systems is fragmentation. Many systems focus on a single capability such as tutoring, assessment, recommendation, or content generation. As a result, the learner may need to move between several unrelated systems during the learning process. Study Buddy addresses this problem by integrating four AI components within one ecosystem: RAG-Based Personalized STEM Storytelling, Multilingual Conversational AI, Emotion-Aware Adaptive Assessment, and Handwritten Content Analysis & Synthesis.

II. RELATED WORK

A. AI-Powered STEM Education

AI has increasingly been applied to STEM education to support personalized learning, adaptive instruction, intelligent tutoring, and automated feedback. Sun *et al.* conducted a systematic review of 33 studies examining AI-supported personalized STEM education in K–12 environments and reported diverse applications across subject knowledge, generic skills, and affective outcomes [1].Leon *et al.* reviewed AI integration across STEM education and identified student agency, adaptive AI-mediated assessment, and ethical and

inclusive design as important themes [2]. Their findings suggest that successful AI education systems should not focus solely on automation but should integrate AI with pedagogical and learner-centered considerations. Villegas-Ch. *et al.* investigated adaptive intelligent tutoring systems for STEM education and reported improved performance through personalized feedback and dynamically adjusted learning experiences [3].Study Buddy extends these approaches by combining personalized generation, conversational tutoring, adaptive assessment, and student-generated content analysis in one ecosystem.

B. Retrieval-Augmented Generation in Education

RAG was introduced as an approach for combining parametric language-model knowledge with retrieved external knowledge [5]. Lewis *et al.* demonstrated that retrieval-augmented architectures are suitable for knowledge-intensive language tasks because retrieved information can provide additional factual grounding. In educational applications, this capability is particularly valuable because generated content should remain aligned with trusted curriculum sources. Study Buddy applies RAG to multiple educational processes rather than using it only for chatbot answering question.The storytelling component retrieves curriculum-aligned information before generating STEM narratives. The conversational component retrieves STEM knowledge before generating tutoring responses. The handwritten-content component retrieves reference information to support verification of student-generated material.

C. Conversational AI for STEM Tutoring

Conversational AI provides learners with immediate access to explanations and can support learning beyond classroom hours. Research on human-AI synergy in STEM education highlights the potential of AI for structured problem solving, personalized support, and real-time feedback while emphasizing that AI should complement rather than replace human teaching [6].Study Buddy extends conventional educational chatbots by combining voice and text interaction, contextual memory, RAG retrieval, visual explanations, multilingual tutoring, voice narration, and personalized recommendations. The project implementation includes Whisper-based speech processing, conversation memory, and frontend-backend integration.

D. Emotion-Aware and Adaptive Assessment

Traditional digital assessments often use fixed question sequences and standardized difficulty. AI-based adaptive education can instead use learner behavior and performance information to modify learning pathways [4]. A recent systematic review covering AI-based adaptive education identifies multimodal learner information and adaptive machine-learning approaches as important directions for personalization. Emotion-aware systems provide an additional source of learner information. Gong demonstrated an educational interaction system combining multimodal emotion recognition and adaptive content delivery, showing the potential of affect-sensitive learning systems [7].Study Buddy incorporates facial-expression information together with response behavior to support learning-state prediction, adaptive feedback, and difficulty adjustment.

E. Handwritten Content Understanding

Handwritten notes contain learner-specific information that is not available to conventional digital learning platforms. OCR and multimodal vision-language models provide a mechanism for converting handwritten content into machine-readable information. When combined with retrieval and language models, this content can be verified, categorized, summarized, and converted into structured revision material. The Study Buddy handwritten-content component extends this approach through a provider-neutral, local-first vision-language pipeline using Ollama with Gemma 4, with Azure GPT-4o available only as an optional fallback. The pipeline incorporates format sniffing, EXIF-orientation correction, auto-contrast, downscaling, and region-aware OCR with per-span confidence and glossary/heuristic correction. Retrieved information is processed using hybrid retrieval combining dense MiniLM embeddings with FAISS, BM25, reciprocal-rank fusion (RRF), and a bge reranker cross-encoder. Evidence-gated verification re-checks cited identifiers in code, uses a domain-restricted web fallback when required, and fails closed when sufficient evidence is unavailable. The verified content is then topic-tagged and synthesized into mind maps, flashcards, and structured notes, while the same pipeline is reused for YouTube transcript analysis.F. Research Gap and Positioning of Study Buddy

The reviewed literature demonstrates strong progress in AI-powered personalization, intelligent tutoring, RAG, affective computing, adaptive assessment, and multimodal educational systems. However, these capabilities are commonly implemented as separate systems.Study Buddy addresses this integration gap through four interconnected components. Its contribution is therefore not limited to introducing another individual AI model. Instead, the system connects multiple AI capabilities into a coherent learning workflow specifically oriented toward the Sri Lankan GCE O/L STEM context.

III. METHODOLOGY

A. Overall System Architecture

Study Buddy STEM AI was implemented as an integrated ecosystem consisting of four major AI components supported by shared knowledge, retrieval, API, machine-learning, and storage services.



Fig. 1. Overall architecture of the Study Buddy STEM AI ecosystem.

The architecture consists of four major components: RAG-based personalized STEM storytelling, multilingual conversational AI, emotion-aware adaptive assessment, and handwritten content analysis and synthesis. A shared knowledge layer contains curriculum resources, NIE textbooks, videos, and other STEM learning materials, while a data and services layer supports retrieval, APIs, machine-learning models, and persistent storage. The architecture shown in the Study Buddy project material explicitly connects the four components with the shared knowledge and service layers.

B. RAG-Based Personalized STEM Storytelling

The first component was developed to address the difficulty of connecting abstract STEM theory with practical and relatable situations. The system begins with learner information and classifies the learner into a suitable persona using a Random Forest model. The persona information is then incorporated into narrative generation. STEM knowledge is retrieved from a semantic vector database containing curriculum-related information.

The system integrates syllabus-anchored RAG retrieval, ChromaDB semantic storage, textbook citation metadata, and chapter-state management to provide reliable learning content. It uses LLM-based narrative generation, Random Forest persona classification, and narrative-to-infographic synthesis to personalize learning. Interactive branching and consequence-based plot generation create adaptive storylines, while diagnostic misconception mapping and mathematical similarity scoring evaluate learner understanding and identify misconceptions. The project specifically defines syllabus anchoring, mathematical similarity scoring, interactive branching, and behavioral verification as major functional requirements.



Fig. 2. RAG-Based Personalized STEM Storytelling workflow

This allows the generated narrative to remain connected to curriculum content while adapting its presentation to the learner.

C. Multilingual Conversational AI

The second component provides an interactive STEM tutor through voice and text. The system receives a student question through text or speech. Speech input is processed using Whisper-based speech recognition. The system maintains conversation history and retrieves relevant STEM information using RAG. The language model then generates a contextual response, which can be supported by visual explanations, voice narration, and personalized recommendations. The implementation includes voice and text interaction, Whisper-based speech-to-text, context tracking, conversation memory, RAG-based STEM retrieval, LLM response generation, visual explanations, multilingual tutoring, personalized recommendations, voice narration, and frontend-backend integration to provide an interactive and personalized learning experience.

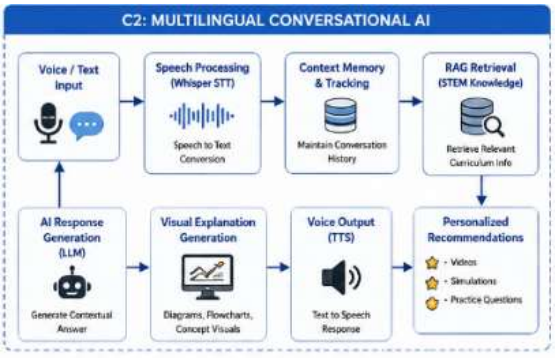


Fig. 3. Multilingual Conversational AI workflow

D. Emotion-Aware Adaptive Assessment

The third component was developed to improve the limitations of fixed-difficulty digital assessments. The system evaluates student responses using multiple signals, including answer correctness, response behavior, response time, answer changes, and facial-expression information. The extracted information is passed to the learning-state prediction component, which supports adaptive feedback and difficulty adjustment.

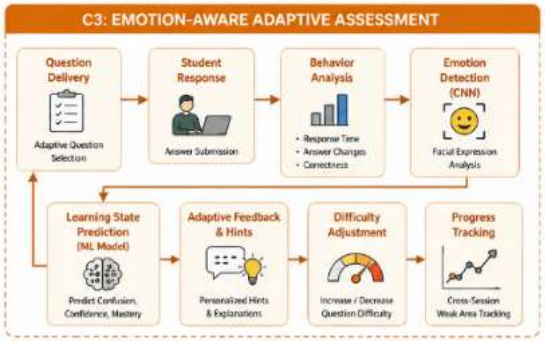


Fig. 4. Emotion-Aware Adaptive Assessment workflow

E. Handwritten Content Analysis and Synthesis

The fourth component, Knowledge Maps, processes student-generated learning material. A student can upload a handwritten STEM note. The system first normalizes the input using format sniffing, EXIForientation correction, auto-contrast, and downscaling, then applies region-aware OCR with per-span confidence and glossary/heuristic correction. A provider-neutral vision LLM is used locally through Ollama (Gemma 4), while Azure GPT-4o is retained as an optional fallback. The extracted content is retrieved against the knowledge base using a hybrid pipeline combining dense MiniLM embeddings with FAISS, BM25,

reciprocal-rank fusion (RRF), and a bge reranker cross-encoder. Evidence-gated verification rechecks cited IDs in code, uses a domain-restricted web fallback when required, and fails closed when evidence is insufficient. Topic extraction provides tags, and the verified content is synthesized into mind maps, flashcards, and structured notes. YouTube transcript analysis reuses the same processing pipeline.

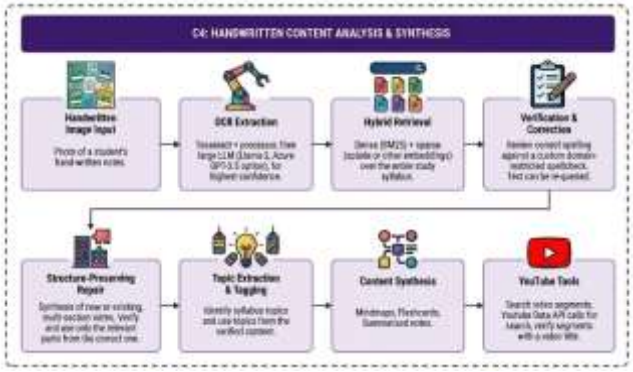


Fig. 5. Handwritten Content Analysis and Synthesis workflow

IV. RESULTS AND DISCUSSION

A. Overall System Implementation

The four components were successfully developed as functional parts of the Study Buddy ecosystem. The project completion documentation reports implementation of the initial RAG pipelines, educational dataset preprocessing, conversational AI prototype, personalized storytelling prototype, emotion-detection integration, adaptive assessment workflow, handwritten OCR processing, knowledge retrieval and verification, database/API development, and initial frontend integration.

B. Personalized STEM Storytelling Results

METRIC DIMENSION	EVALUATED TARGET	RESULT / BENCHMARK VALUE	STATUS ASSESSMENT
Generation Quality Score	Multi-rubric educational quality score (0–100)	91.7% (Range: 83.3% – 100.0%)	High Quality
Persona Theme Alignment	Predicted theme matching gold-standard theme	66.7% (8/12 matches)	Needs Calibration
JSON Output Conformity	Schema parsing & strictly valid JSON returns	100.0% (12/12 successful)	100% Compliant
RAG Semantic Retrieval	Best similarity match against source textbooks	78.0% (Range: 71.8% – 81.9%)	Optimal Grounding
Narrative Generation Volume	Target ~200–300 words with 3-paragraph structure	230.9 words (100% 3-para pass)	Target Met
Execution Latency	Total wall-clock generation runtime	43.8 sec/case (Total: 525.1s)	Standard Load

Fig. 6. Quantitative Evaluation Of The Personalized Stem Storytelling Component

The personalized STEM storytelling component was evaluated across six implementation and generation criteria: generation quality, persona-theme alignment, JSON output conformity, RAG semantic retrieval, narrative generation volume, and execution latency. The evaluation used predefined targets and benchmark values to assess the reliability and consistency of the storytelling pipeline.The component achieved a 91.7% generation quality score, with individual rubric scores ranging from 83.3% to 100.0%, indicating strong overall educational content quality. JSON output conformity reached 100.0% (12/12 successful cases), demonstrating that the generated responses consistently satisfied the required structured-output schema. The RAG retrieval component achieved a 78.0% semantic similarity score, with observed values ranging from 71.8% to 81.9%, indicating effective grounding against the source textbook material.

The narrative-generation component produced an average of 230.9 words per case, satisfying the target of approximately 200–300 words with a three-paragraph structure. The measured execution latency was 43.8 s per case, with a total wall-clock runtime of 525.1 s for the evaluated cases. Persona-theme alignment achieved 66.7% (8/12 matches), identifying persona consistency as the main area requiring further calibration.

Overall, the results demonstrate strong generation quality, structured-output compliance, curriculum grounding, and target narrative-length adherence. However, the lower persona-theme alignment indicates that further refinement of the persona classification and theme-matching process is necessary. The evaluation therefore supports the feasibility of the personalized storytelling pipeline while also identifying a specific area for future optimization.

C. Multilingual Conversational AI Results

The multilingual conversational AI component was evaluated through a comparative assessment of three language models—Mistral 7B, Qwen2.5 7B, and Llama 3.2 3B—across five criteria: response quality, context relevance, explanation quality, response speed, and resource efficiency. The overall scores were 82.6, 88.6, and 83.4, respectively.

Qwen2.5 7B achieved the highest overall score of 88.6, demonstrating a balanced performance across response quality, contextual relevance, explanation quality, response speed, and resource efficiency. It achieved scores of 89 for response quality, 90 for context relevance, 90 for explanation quality, 88 for response speed, and 86 for resource efficiency. Mistral 7B achieved higher scores in response quality (92), context relevance (91), and explanation quality (93), but its lower response speed (65) and resource efficiency (72) reduced its overall score to 82.6. Llama 3.2 3B demonstrated the strongest response speed (95) and resource efficiency (94), but obtained comparatively lower scores for response quality (78), context relevance (76), and explanation quality (74), resulting in an overall score of 83.4.

Based on the comparative evaluation, Qwen2.5 7B was selected as the preferred conversational model because it provided the highest overall score and a balanced trade-off between response quality, contextual relevance, explanation quality, response speed, and resource efficiency. The results support the suitability of the selected model for contextual STEM tutoring within the Study Buddy

ecosystem.

MODEL COMPARISON RESULTS

Metric	Mistral 7B	Qwen2.5 7B	Llama 3.2 3B
Response Quality	82	89	78
Context Relevance	81	88	76
Explanation Quality	93	90	74
Response Speed	65	88	85
Resource Efficiency	72	86	94
Overall Score	**82.6**	**88.6**	**83.4**

Fig. 7. Comparative Evaluation Of Conversational Ai Models

D. Emotion-Aware Adaptive Assessment Results

The emotion-aware adaptive assessment component was evaluated in terms of the computational latency of the facial-expression detection models used within the assessment pipeline. Two YOLO-based configurations, YOLOv8s and YOLOv8m, were compared using CPU latency per frame. As shown in Fig. X, the selected YOLOv8s configuration achieved a latency of 27.7 ms per frame, whereas the rejected YOLOv8m configuration required 59.6 ms per frame. The selected configuration therefore reduced CPU processing latency by 31.9 ms per frame, corresponding to an approximately 53.5% reduction relative to YOLOv8m.

The lower latency of YOLOv8s provides a practical advantage for real-time emotion-aware assessment, where facial-expression information must be processed alongside learner responses and behavioral signals. The selected detector can process approximately 36.1 frames per second based on the measured CPU latency, compared with approximately 16.8 frames per second for YOLOv8m. Therefore, YOLOv8s was selected as the more suitable configuration for the assessment pipeline because it provides substantially lower computational latency while supporting real-time facial-expression processing.

The detected emotion information is subsequently combined with assessment-related signals, including answer correctness, response time, and answer changes, to construct a learner-state representation. This representation supports adaptive feedback and difficulty adjustment rather than relying solely on fixed assessment sequences.

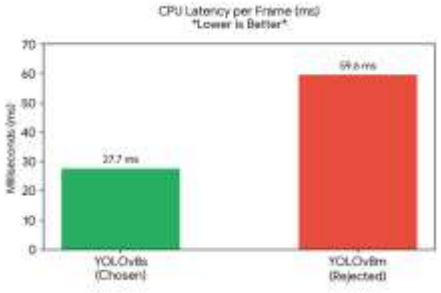


Fig. 8. CPU latency comparison of YOLOv8 configurations for facial-expression detection.

E. Handwritten Content Analysis Results

The handwritten content analysis component was evaluated using a comparative demonstration profile across eight project criteria: text understanding, information extraction, OCR noise resilience, reasoning ability, output structure, consistency, inference speed, and deployment ease. Three vision-language models were compared: Gemma 4 Cloud, Llama 3.2, and Qwen 2.5. As shown in Fig. X, Gemma 4 achieved the highest overall demonstration rating, with an average score of 88.6/100, compared with 85.5/100 for Qwen 2.5 and 82.4/100 for Llama 3.2. Gemma 4 achieved its highest scores in deployment ease (95), output structure (92), and consistency (90), while obtaining 90 for text understanding. It also demonstrated strong performance in information extraction (85), OCR noise resilience (85), reasoning ability (87), and inference speed (85). Qwen 2.5 showed competitive performance, particularly in consistency (88) and inference speed (88), whereas Llama 3.2 achieved its strongest results in inference speed (92) and deployment ease (90). Based on these demonstration results, Gemma 4 was selected as the preferred model for the local-first Knowledge Maps implementation.

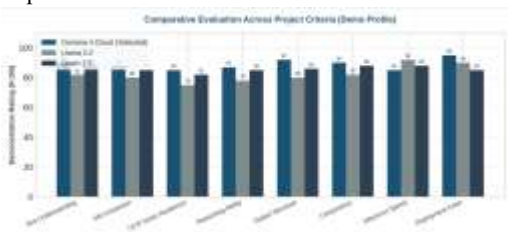


Fig. 9. Comparative evaluation of vision-language models for Knowledge Maps.

F. Integrated Learning Workflow

The main strength of Study Buddy is the integration of the four components. A student can first learn a STEM concept through personalized storytelling, ask follow-up questions through the conversational tutor, demonstrate understanding through adaptive assessment, and finally upload handwritten notes for verification and synthesis. This integration transforms the system from a collection of independent AI tools into a continuous learning ecosystem.

V. DISCUSSION AND LIMITATIONS

The successful implementation of Study Buddy demonstrates that multiple AI technologies can be combined to support different stages of the STEM learning process. The storytelling component addresses engagement and conceptual understanding by presenting STEM concepts through personalized narratives. The conversational component provides immediate tutoring and supports multimodal interaction. The adaptive assessment component introduces behavioral and affective information into assessment decisions. The handwritten-content component enables student-generated learning material to be processed and reused. The system is particularly relevant to Sri Lankan O/L education because it is grounded in curriculum-oriented resources and addresses multilingual and handwritten-learning requirements. The project specifically shifted toward an O/L focus and replaced an earlier augmented-reality direction with RAG-based approaches to improve practical accessibility.

VI. CONCLUSION

This paper presented Study Buddy STEM AI, an integrated AI-powered STEM learning ecosystem developed for personalized, adaptive, and multimodal education in the Sri Lankan GCE O/L context. The completed system integrates four AI components: RAG-Based Personalized STEM Storytelling, Multilingual Conversational AI, Emotion-Aware Adaptive Assessment, and Handwritten Content Analysis & Synthesis. The system provides a strong technical foundation for AI-supported STEM learning at the GCE O/L level and demonstrates the practical potential of combining RAG, generative AI, machine learning, affective computing, speech processing, OCR, and knowledge synthesis in a single educational platform. Further large-scale learner evaluation will establish the educational effectiveness and generalizability of the completed system.

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